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Automatic recognition of strawberry diseases and pests using convolutional neural network

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ABSTRACT

In this study, convolutional neural network (CNN) was used to classify and recognize the strawberry diseases and pests which cause tremendous economic losses for farmers. Based on AlexNet, the goal of this paper is to improve the recognition algorithm of strawberry diseases and pests by using the following methods: First, transfer learning is used to accelerate the time of model training and improve the recognition accuracy. Second, the model is fine-tuned to reduce the amount of training parameters and training time. The operator of combing inner product and ℓ_∞ -norm was proposed and applied in the last fully connected layers to further improve the accuracy of recognition for strawberry diseases and pests. Compared with other start-of-the-art models, the experimental results show that our model achieves lower computational complexity and higher top-1 accuracy of the recognition for the strawberry diseases and pests on the test set.

1. Introduction

In recent years, with the development of computer science and technology, machine learning, especially deep learning, has been numerous applied in the field of life [6,11,18,20]. Besides, CNN, one of the deep learning methods, has achieved excellent performance in a sea of tasks. AlexNet [12] is a classical convolutional neural network model which used widely by many scholars. For example, Lu et al. [14] introduced a method to automatically classify and detect the sounds of long-fined pilot whales, killer whales and harp seals based on AlexNet with transfer learning. And all experimental results showed that the proposed method had outstanding performance and immense potential for practical application. M. He et al. [8] proposed a skeletal maturity assessments method, and the experimental results showed that the bone age evaluation based on the improved AlexNet model proposed on the research was much more accurate. Dhar et al. [5] introduced a universal method which utilized the AlexNet model to detect abnormal heart sounds. A pre-trained AlexNet model had been used and fine-tuned in order to improve the system performance. Compared with other classical classifiers, the proposed approach was outperformed by experimental results. And the results of the study showed that their techniques attained an accuracy of 98%. Chen et al. [4] proposed a 3D AlexNet methodology for automatic segmentation of prostate cancer magnetic resonance images. The improved 3D AlexNet can automatically complete the structured segmentation of prostate resonance images. Compared with other algorithms, the performance of the 3D AlexNet was excellent in terms of parameter amount, training time and network performance evaluation. Using an improved AlexNet model, Ju and Xue [9] aimed at improve the fish recognition by self-attention mechanism and transfer learning. The experimental results showed that the proposed model achieved less computational complexity and higher accuracy compared to other state-of-the-art fish recognition methods. In the field of agriculture, computer technology is also utilized widely. For instance, Juan Carlos et al. [10] was aimed at strengthening the resources that farmers had for crops management of pests and diseases in a digital way using Internet of things which can obtain crucial information of variables within a strawberry crop such as temperature and pH. Pérez-Borrero et al. [15] proposed a method for instance segmentation of strawberries based on deep learning techniques. In this study, a new dataset of 3100 images was used to train the neural network that they designed. The experimental results on the test set showed that the proposed method could be regarded as the most promising one in view of its possible integration in automatic strawberry picking systems.

As we all know, because strawberry is rich in nutritional value [1,17,21], strawberry is regarded as a prominent fruit for its wonderful and flavor properties and it plays a key role in our daily life. In addition, strawberry is also a crucial cash fruit which can bring huge economic benefits for farmers in the north of China [2,3,19]. However, there were enormous economic losses caused by strawberry diseases and pests every year. If strawberry diseases and pests can't be recognized correctly or timely, the farmers, especially for those who don't own rich

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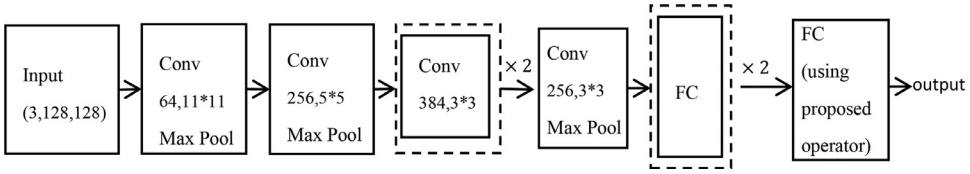


Fig. 1. The structure of our CNN model (Conv: convolutional layer, FC: fully connected layer).

experience in planting, will not be able to spray pesticides correctly to prevent and control diseases and pests. Furthermore, spray pesticides incorrectly may cause excessive pesticide residues and more economic losses. Although deep learning was used widely in the agriculture field, there is little research which utilized CNN to recognize multi-category strawberry diseases and pests. In order to help farmers to recognize the diseases and pests correctly and timely, a model that has high accuracy and low memory requires is designed based on the AlexNet in this paper.

2. The related theory

2.1. The structure of CNN

CNN is a highly effective method that was used widely in the domain of image classification and recognition. The traditional CNN model is usually composed of convolutional layers, pooling layers and fully connected layers. Firstly, the convolutional layers are utilized to extract features from images, then put the feature map into pooling layers which can reduce the training parameters and prevent over-fitting. In addition, the fully connected layers are utilized to map the features into non-linear space and classify the feature to a category that took a maximum probability. The structure of neural network that we used is shown in Fig. 1.

2.2. Inner product and our proposed operator

In the hidden layer of CNN, the output value is calculated by the inner product of the feature matrix and the weight matrix, then the output value will be activated through an activation function, like Relu or sigmoid function.

Suppose there are N samples which are fed to a hidden layer that includes L neural nodes. And there are M features in every sample, so the output matrix Y ($Y \in R^{N\times L}$) will be calculated by the dot product of the input matrix X ($X \in R^{N\times M}$) and the matrix of weight values W ($W \in R^{L\times M}$).

Mathematically, using the inner product, the ith output value about the kth sample can be expressed as Eq. (1).

$$y_i^k = \sum_{j=1}^{M} X_{kj} * W_{ji}^T, i = 1, 2, ..., L. \tag{1}$$

And the output value Y for one hidden layer can be given as

$$Y_{ij} = \sum_{a=1}^{M} X_{ia} * W_{aj}^T \tag{2}$$

where Y_{ij} is the value of the ith row and the jth column of the output matrix Y, and X_{ia} is the ath value of the ith row of the matrix of input X, and W_{aj}^T is the jth value of the ath column of the matrix of W^T. And $\sigma(\cdot)$ represents activation function, such as sigmoid or Relu activation function. Eqs. 2–5

In our proposed operator, combining the inner product with ℓ_∞-norm, the improved operator is expressed as follows

$$\widetilde{y}_i^k = \frac{\sum_{j=1}^{M} X_{kj} * W_{ji}^T}{\left\| W_i^T \right\|_\infty}, i = 1, 2, \cdots, L, \tag{3}$$

where W_i^T is the ith column of the weight matrix W^T and $\tilde{y}_i^k$ is the ith output value that for the kth sample. And the output value of all samples through the hidden layer can be expressed as

$$\widetilde{Y}_{ij} = \frac{\sum_{a=1}^{M} X_{ia} * W_{aj}^T}{\left\| W_i^T \right\|_\infty} \tag{4}$$

where $\tilde{Y}_{ij}$ is the value of the ith row and the jth column of the output value matrix $\tilde{Y}$(that is generated by input values and weight values using our proposed operator).

3. Experiments and data

3.1. Data processing

For the sake of training a model that can be used to recognize and classify the strawberry diseases and pests, the images data were collected using the smartphone camera by us in the cultivated land of real world. Next, the images of strawberry diseases and pests were annotated with the corresponding categories manually with the help of farmers which owned the rich experience in strawberry planting. In addition, to make the model more robust instead of overfitting on the training dataset, data augmentation was utilized on our dataset, including changed the brightness, randomly cropped, etc. After data augmentation, there are 6608 images in total which are split into the training set (that occupied 80% of images data) and the test set (that occupied 20% of images data). The dimension of each strawberry image is (128, 128, 3) and there are 9 different categories of strawberry diseases and pests in the dataset, including virus disease, root rot, powdery mildew, thrips, botrytis cinerea, sunburn, vegetables green insect, slug and aphid. Some examples of our data are shown in Fig. 2.

3.2. Details of model

Based on a light-weight network, the classical AlexNet model, some improvements have been achieved, and then the improved AlexNet is adopted as our model which will be deployed to the mobile device to help farmers who want to recognize the strawberry diseases and pests conveniently and quickly in the future. First, the AlexNet neural network was fine-tuned for the sake of reducing the parameters and adapting the size of our images data. In addition, we utilized the pre-trained parameters that the AlexNet model had been trained on the ImageNet dataset. Finally, the inner product was replaced with our proposed improved operator in the last fully connected layer which was used to classify the features. The details of our model are shown in Table 1.

3.3. Training of model

First, we utilized the pre-trained parameters that the original AlexNet model was trained on the ImageNet dataset. In other words, according to the method of transfer learning, the parameters in convolutional layers were frozen and only the fully connected layers' parameters were updated in the process of training our improved AlexNet model. Secondly, the Adam algorithm was utilized to update the parameters and accelerated the training process. Moreover, multi-class cross entropy loss function was used in experiments. In addition, the learning rate was set to 0.001 and 0.0001 respectively and the batch size was set to 2^4, 2^5, 2^6, 2^7 and 2^8 respectively in experiments. Finally, by combing different learning rate and batch size, the model which attained the highest top-1



Fig. 2. Some examples of our image dataset.

Table 1
Details of various layers of our model.

Layers	Output Size	param
Input data	3×128×128	-
Convolutional Layer 1	64×31×31	23,296
Relu	64×31×31	0
Max Pooling Layer 1	64×15×15	0
Convolutional Layer 2	256×15×15	307,392
Relu	256×15×15	0
Max Pooling Layer 2	256×7×7	0
Convolutional Layer 3	384×7×7	663,936
Relu	384×7×7	0
Convolutional Layer 4	384×7×7	884,992
Relu	384×7×7	0
Convolutional Layer 5	256×7×7	590,080
Relu	256×7×7	0
Max Pooling Layer 5	256×3×3	0
Dropout	2304	0
Linear layer	512	1,180,160
Relu	512	0
Dropout	512	0
Linear layer	128	65,664
Relu	128	0
Linear layer (using proposed operator)	9	1,161

accuracy for classification and recognition of strawberry diseases and pests was picked up and saved after 128 iterations of epoch.

All experiments were performed on the Tesla P100-PCIE GPU (12GB RAM) under the operating system of CentOs 7. The code was implemented using Pytorch1.8 library in python 3.8.5, and the code was available at https://gitee.com/dc2019/improved_alexnet3.

Table 2
The path from AlexNet to our improved AlexNet.

	AlexNet			Our improved AlexNet
Transfer learning?		√	√	√
Fine-tuned?			√	√
Improved operator?				√
Top-1 accuracy	93.80%	96.29%	97.05%	97.35%

4. Experimental results and discussion

Compared with AlexNet model, the paths of improving accuracy were showed in Table 1 after applying some tricks that we used in our improved AlexNet neural networks. From Table 2, we can see that the top-1 accuracy of classification and recognition for strawberry diseases and pests has been improved in varying degrees. By applying the method of transfer learning, the accuracy had been taken a significant improvement, and then by using the proposed operator and fine-tuned the model, the top-1 accuracy was further improved.

In addition, the details of parameters about models that we utilized in experiences are shown in Table 3, and the corresponding results are shown in Table 4. By experimental results, we compared our model with other methods in some metrics, such as top-1 accuracy, model size and cost time in the training process. The equation of accuracy is shown as

Table 3
The details of parameters about models.

Models	Total params	Trainable params	Non-trainable params	Forward/backwardpass size (MB)
AlexNet	28,894,409	28,894,409	0	3.16
Resnet50	23,526,473	23,526,473	0	93.58
MobileNet_v2	2,235,401	2,235,401	0	49.92
SCNet50	23,534,025	23,534,025	0	100.49
Our model	3,716,681	1,246,985	2,469,696	2.50

Table 4
Comparison of classification results on test dataset.

Models	Accuracy	Average training time (s)	Model size (MB)
AlexNet	93.80%	652.7	110
Resnet50	96.44%	2310.1	90.0
MobileNet_v2	93.49%	1276.2	**8.53**
SCNet-50	95.70%	3384.5	90.2
Our model	**97.35%**	**363.6**	14.1

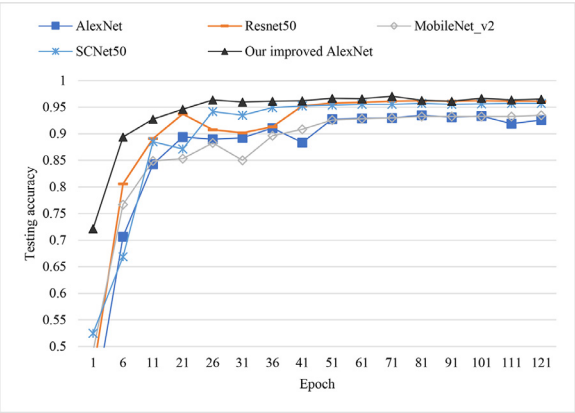


Fig. 3. Testing accuracy of different models.

$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

where TP is true positive, TN is true positive, FN is false negative and FP is false positive.

From Tables 2 and 3 we can see clearly that our model achieves a good performance which the top-1 accuracy has increased from 93.80% to 97.35% and the model size has reduced from 110 MB to 14.1 MB based on AlexNet networks. Although our model size is slightly bigger than MobileNet_v2 [16], our model consumes less training time and has higher top-1 accuracy for classification and recognition of strawberry diseases and pests by Table 4.

The reason why our model consumed less training time than MobileNet_v2, Resnet50 (K. [7]) and SCNet50 [13] was that our model has so few convolutional layers and trainable parameters that the parameters can iterate rapidly by transfer learning. In addition, comparing our model with AlexNet model, our model took less time on the process of training due to the neural nodes in fully connected layers has reduced a lot by fine-tuned the model. In addition, the neural nodes in fully connected layers occupy abundant memory, so the model size is reduced a lot after cutting out neural nodes. Besides, the accuracy of recognition and classification is increased after using the proposed operator because the proposed operator can measure the output feature better. Moreover, from Figs. 3 and 4 we can also conclude that our model keeps higher accuracy and lower loss than other models.

Furthermore, because of using transfer learning, our model keeps fast convergence speed so that it gets a higher accuracy on the test set at the beginning of training process. Using transfer learning, not only the convergence speed is promoted, but also the problem of insufficient training data can be effectively overcome so that the accuracy of classification is improved. From Fig. 5, we can draw that our model also performs well in most classes although the accuracy of classification is slightly lower than other methods. The accuracy of classification for all categories has been further promoted on the basis of AlexNet model. And compared with other models, our model also achieves a higher accuracy on the test set.

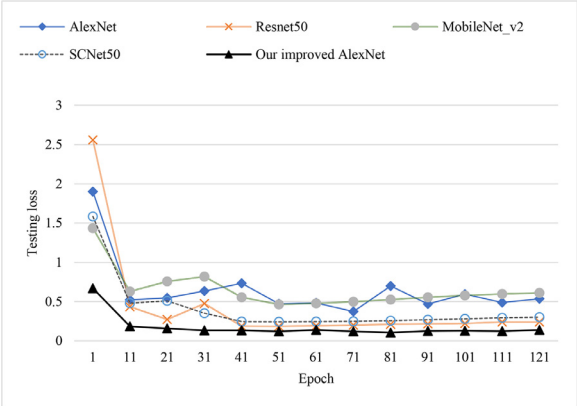


Fig. 4. Testing loss of different models.

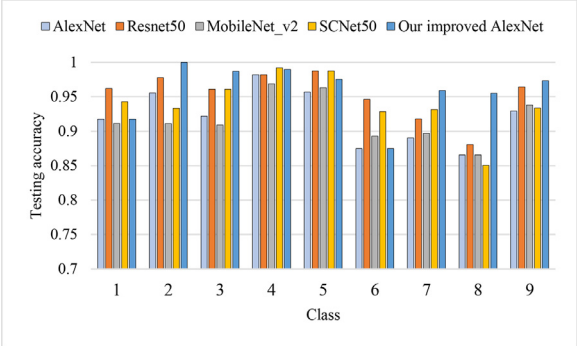


Fig. 5. Top-1 accuracy comparison of different classes (The number 1 to 9 represent different classes of strawberry diseases and pests: powdery mildew, virus disease, vegetables green insect, root rot, botrytis cinerea, thrips, sunburn, slug and aphid).

5. Conclusion

In this paper, an improved AlexNet model was designed for the classification and recognition of strawberry diseases and pests. Firstly, transfer learning was utilized to overcome the problem of over-fitting in the process of training neural networks when the training data was insufficient. Besides, we fine-tuned the model to promote the accuracy and reduce the model parameters. High accuracy will help farmers make a better adjustment for spraying pesticides when strawberries are infected with diseases and pests, and fewer parameters will reduce the model size

for the sake of accelerating the time of reasoning on image and reducing the internal storage when the model is employed on mobile devices. In addition, combining inner product with ℓ_∞-norm, we proposed an improved operator that can be applied to the last fully connected layers of neural networks. By using the improved operator, the accuracy of classification had been improved further in most cases. The models were trained on our collected images data in real-life, and the experimental results showed that our method performed better than other start-of-the-art methods for the classification of strawberry diseases and pests.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests that could have appeared to influence the work reported in this paper.

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